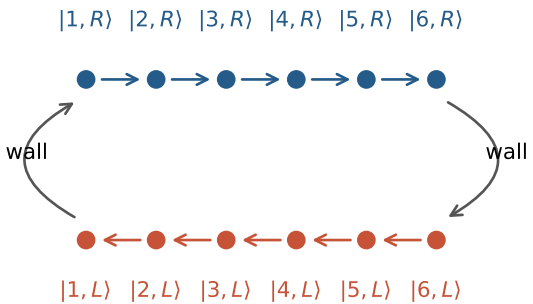
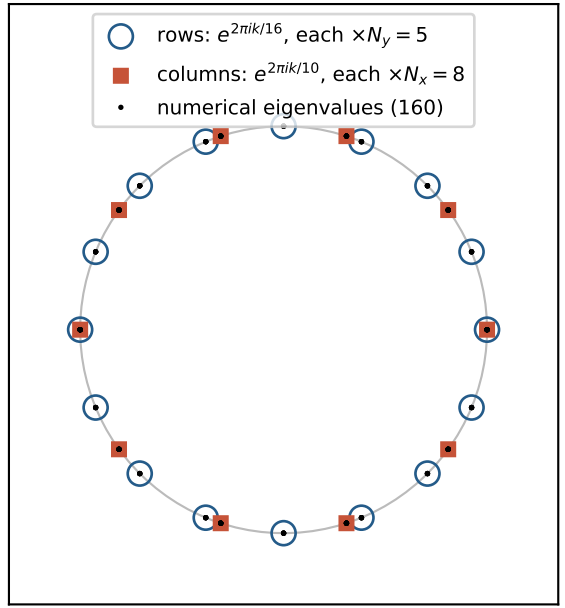
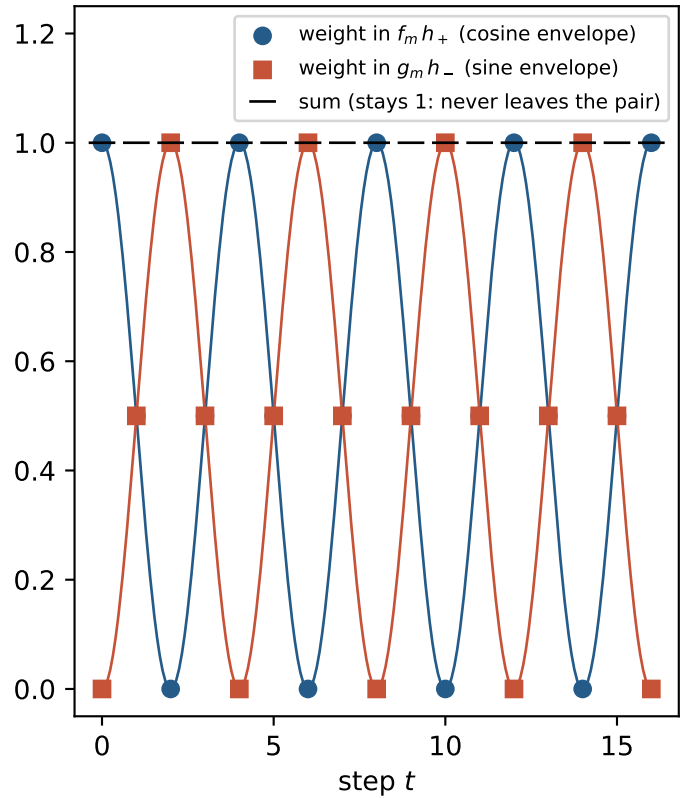
(a) One row ($N_x = 6$): a single cycle of length $2N_x = 12$



With $C = I$ the walk is the shift alone: every row is one closed cycle of $2N_x$ basis states, every column one of $2N_y$. A state never leaves its cycle.

(b) Spectrum of $U = S$ on 8×5: roots of unity



(c) S on one row rotates $f_m h_+$ into $g_m h_-$ ($N_x = 8$, $m = 2$, angle $\pi m/N_x$ per step)